10.7 A sustainable energy future

The production of energy from different resources has both positive and negative impacts for the environment, for people, and for the economy. Governments and policy makers need to take these impacts into consideration when they are determining the energy mix for a country or region.

Positive impacts

A secure energy supply offers economic and social benefits. Economically, a secure supply of energy allows for the development of commercial farming, industry, and businesses within a country. This encourages economic stability and allows for ongoing industrial and economic expansion.

From this economic stability comes social development, with access to education, medical care, and jobs improving the quality of life and standard of living of the population. The production of energy, for example the mining, processing, and production of energy, also creates jobs. This increases a population's chance for a better standard of living.

Renewable energy resources have additional positive social and environmental impacts to non-renewable resources. For example, they do not emit greenhouse gases, not only reducing the impact on the climate, but also reducing the risk of illness and lung disease.

Negative impacts

Although we can categorise the negative impacts of different sources of energy into economic, social, and environmental categories, it is important to recognise that many of these impacts are interconnected.

Environmental impacts

Table 10.4: Environmental impacts of using different energy sources

Non-renewable	Renewable
During mining for coal, oil, natural gas, and uranium, environmental damage occurs, with loss of habitat, reduced biodiversity, and the potential for pollution due to events like oil spills.	Any materials that are used in the construction of renewable energy infrastructure (for example, wind turbines, dam walls) all have an environmental footprint.
Burning of fossil fuels, especially coal, releases sulphur dioxide into the atmosphere which causes acid rain. Acid rain is harmful to ecosystems, causing plants and aquatic animals to die.	Wind turbines can injure or kill birds which fly into the blades.
The burning of fossil fuels creates poor air conditions and can result in the formation of smog over cities. This reduces visibility and causes harm to living organisms.	Areas of natural habitat may have to be cleared for installing power plants, such as large solar farms.
Burning fossil fuels releases greenhouse gases such as carbon dioxide, which cause pollution of the atmosphere, increasing the rate of climate change and causing global warming.	
Nuclear power uses radioactive material which is harmful to life if it is not correctly managed. The long-term impact to an ecosystem can be devastating.	

Economic impacts

Table 10.5: Economic impacts of using different energy sources

Non-renewable	Renewable
Cleaning up an oil spill or rehabilitating an old mine site is expensive.	The loss of jobs in the non-renewable energy sector is a concern to many countries who have developed their economy around fossil fuels.
Acid rain damages buildings, eroding the stonework and causing it to crumble. This requires money to continuously repair or slow the damage.	Switching to renewable energy has a significant financial cost. This is challenging for both LICs and HICs.
Climate change is causing increasing costs around the world. Rising temperatures and sea levels, and changes to precipitation rates all create costs as countries adapt to these changes.	
A nuclear accident shuts down regions around a nuclear plant and destroys the economy in the area reliant on that energy source.	

Social impacts

Table 10.6: Social impacts of using different energy sources

Non-renewable	Renewable
Air pollution from the burning of fossil fuels contributes to health conditions, such as asthma, bronchitis, lung cancers, high blood pressure, and heart conditions.	Wind turbines and solar panels may be viewed as unattractive.
Acid rain is an irritant, causing skin conditions. It also causes crops to fail, resulting in food insecurity.	Wind turbines can be noisy, disturbing local populations.
Smog is toxic and causes harm to the skin, eyes, and lungs.	The loss of jobs in regions where the extraction or use of non-renewable energy resources is the main employer decreases quality of life and increases poverty.
Climate change causes social anxiety, where people worry about how climate change will affect their future. Changes to the climate puts pressure on economies, resulting in the loss of jobs and a reduced quality of life. Changes to climate also impact food supply, putting pressure on food security and impacting the health of the population.	
In the event of a nuclear accident, the exposure of people to radioactive material may result in burns, cancers, and death.	

TASK 1

a Create a spider diagram that shows how the environmental, social, and economic impacts of different energy sources are interconnected. Look for what impacts occur and how they relate to each category.

b In pairs, carry out research to identify three cities around the world that suffer from high levels of smog. Give reasons for why these particular cities suffer from smog and then compare your findings with others in your class.

REFLECTION

How did you find creating a spider diagram that links different concepts that have shared impacts? Did you remember seeing a similar diagram in Topic 4 and use it as a guide in your thought process? Do you find the creation of this type of image helps you to remember new information?

Health hazards caused by coal power stations

A country that currently produces most of its electricity from coal power is looking at expanding the number of coal-fired power stations. The long-term pollution caused by these power stations are a burden on the healthcare system. The air pollution that the power stations cause results in the death of approximately 5000 people a year and leads to almost 3000 early births of babies each year. Almost 27,000 cases of bronchitis in children are caused by the pollution from the power plants every year, and as many as 1.48 million working days are lost each year due to the population being ill.

Air pollution from these power stations leads to heart and lung disease and increases the risk of many other types of diseases. The government has found that almost half of the €10.9 billion spent on health care each year is spent on people living near the coal power stations

Source A: Newspaper article about coal power stations

TASK 2: Study Source A

a In pairs, read through the article on the health hazards caused by coal power stations. Then discuss if you think moving towards cleaner energy sources would be better for:

- **i** The health of the population
- **ii** The health of the environment
- **iii** The stability of the economy.

b Write a 200 to 300-word letter to the government motivating for a change in the source of energy in the country.

Managing energy supplies

There are a number of strategies and techniques that we can use to manage energy to improve sustainability. Some important strategies include energy conservation, increasing energy efficiency, changing the energy mix, and developing political policies. Being more conscious of our energy use reduces the pressure on having to provide a limitless supply of energy.

When evaluating the sustainability of an energy source, we need to consider how countries, companies or people are working towards cleaner energy solutions while reducing their energy consumption. When energy projects have sustainable goals, they plan to meet very specific targets, all of which work towards a cleaner and more efficient energy solution designed to supply energy while protecting the environment. These sustainable goals should include long term strategies such as plans and policies to meet energy needs, as well as techniques that can be used to make those strategies successful. Techniques include factors such as the education of the population and the development of technology that can help to conserve energy and increase energy efficiency.

Energy conservation

Energy conservation involves using less energy through the practice of simply using less energy! At an individual level, this involves ideas like turning off lights when not in a room, unplugging appliances when not in use, and walking or cycling instead of driving. However, it also includes ideas such as the insulation of houses to control inside temperatures and reduce the need for heating or cooling of buildings.

Increased energy efficiency

Improving energy efficiency is the process of using less energy to get the same task done. This is done through energy conservation and the development of technology that uses less energy. For example, washing machines, fridges, lights or vehicles that use less energy but do the same job. In addition, new computerised smart homes or systems are able to lower energy consumption by identifying areas where equipment can be switched off.

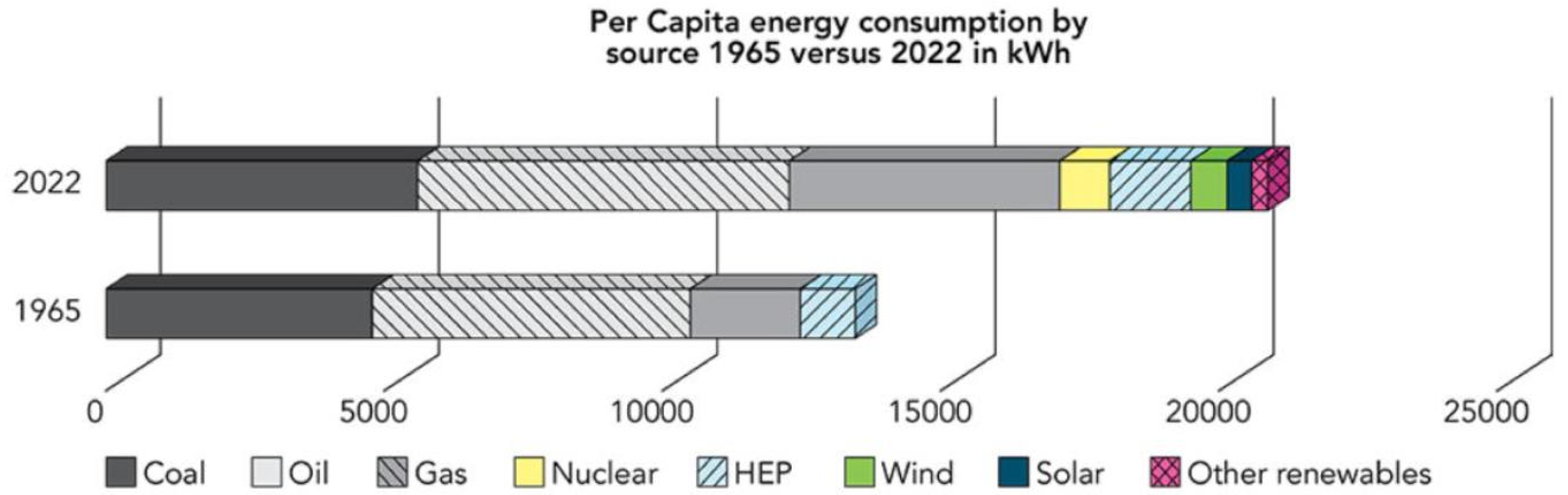
The energy mix

Increasing the variety of energy sources a country relies upon improves the potential to supply energy to the population. When a country or region is reliant on only one energy source, there is the risk of that energy supply being cut off. For example, if a country only uses coal to produce electricity, then if there is a problem with the coal supply, electricity production will stop. However, if the mix included solar, wind, and HEP, then when the coal supply was limited, other energy sources could be used.

Political policies

Governments need to consider a wide range of areas when looking at how to change to more sustainable energy and green energy policies. They need to consider the current energy requirements of the country, including seasonal variations in energy consumption, and determine which energy sources are likely to supply energy that meets their demand. For example, is there sufficient space and wind for the installation of wind turbines, or is there an opportunity to rely more on solar energy?

The success of implementing strategies and techniques in a country depends on a wide variety of factors. Firstly, the economic and political stability of a country, which helps to ensure that finances are available to change from one energy source to another or use a mix of energy sources. In addition, there needs to be the political willingness to develop energy strategies and then put them in place. Secondly the country may not have access to a wide variety of energy sources, making the change from one to another challenging.



Source B: Global energy consumption per capita by resource in 1965 and 2022

TASK 3: Study Source B

a **Which non-renewable resource** showed the largest increase between 1965 and 2022?

b **Which renewable energy resource** was already being used in 1965?

c In 1965, the total estimated energy consumption per capita (per person) was 12975 kWh, while in 2022 it was 20889 kWh.

i **Calculate the difference** in energy consumption per capita between 1965 and 2022.

ii Suggest why energy consumption per capita has increased so significantly between 1965 and 2022.

d **Describe and explain the change** in energy mix between 1965 and 2022.